

# Jeonghoon Lim

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🌐 <https://jhlim0918.github.io/>

📄 ADS

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## Education

- 2020-present 📖 **Iowa State University, Ames, IA**  
Ph.D. in astrophysics: *Simulating the Streaming Instability and Planetesimal Formation*; Advisor: Prof. Jacob B. Simon
- 2012-2020 📖 **Chungnam National University, Daejeon, South Korea**  
M.S. in astronomy: *Statistics of Turbulence Driven by Solenoidal and Compressive Drivings*; Advisor: Prof. Jungyeon Cho  
B.S. in astronomy

## Research Interests

- 📖 Planet formation; Protoplanetary disks; Magnetohydrodynamics; Computational astrophysics

## Awards and Achievements

- 2022-2025 📖 Future Investigators in NASA Earth and Space Science and Technology (FINESST)
- 2020-2022 📖 The Korean Government Scholarship Program for Study Overseas.
- 2020 📖 Graduate Research Award by Journal of the Korean Astronomical Society.

## Computational Skills/Experience

- 📖 Godunov Schemes; Particle-mesh; Grid codes
- 📖 High-performance computing on supercomputers in ACCESS and NASA-HEC
- 📖 Co-PI on 2022 – 2023 ACCESS Allocation (AST120062, 29 Million CPU Hours, PI: J. Simon, “Numerical Simulations of Planetesimal Formation”)
- 📖 Codes/Programming Languages: Athena, Julia, C, Python

## Publications

### Journal Articles

- 1 D. Carrera, L. E. J. Eriksson, J. **Lim**, W. Lyra, and J. B. Simon, “Positive feedback: II. How dust coagulation inside vortices can form planetesimals at low metallicity,” *A&A*, vol. 701, L1, p. L1, Sep. 2025. 🌐 DOI: 10.1051/0004-6361/202555045.
- 2 D. Carrera, J. **Lim**, L. E. J. Eriksson, W. Lyra, and J. B. Simon, “Positive feedback: How a synergy between the streaming instability and dust coagulation forms planetesimals,” *A&A*, vol. 696, L23, p. L23, Apr. 2025. 🌐 DOI: 10.1051/0004-6361/202554100.
- 3 J. **Lim**, S. A. Baronett, J. B. Simon, *et al.*, “Bridging Unstratified and Stratified Simulations of the Streaming Instability for  $s = 0.1$  Grains,” *arXiv e-prints*, arXiv:2505.23902, arXiv:2505.23902, May 2025. 🌐 DOI: 10.48550/arXiv.2505.23902.
- 4 J. **Lim**, J. B. Simon, R. Li, O. Brouillette, D. G. Rea, and W. Lyra, “The Streaming Instability in 3D: Conditions for Strong Clumping,” *arXiv e-prints*, arXiv:2509.18270, arXiv:2509.18270, Sep. 2025. arXiv: 2509.18270 [astro-ph.EP].

- 5 J. **Lim**, J. B. Simon, R. Li, *et al.*, “Probing Conditions for Strong Clumping by the Streaming Instability: Small Dust Grains and Low Dust-to-gas Density Ratio,” *ApJ*, vol. 981, no. 2, 160, p. 160, Mar. 2025.  DOI: 10.3847/1538-4357/adb311.
- 6 J. **Lim**, J. B. Simon, R. Li, *et al.*, “Streaming Instability and Turbulence: Conditions for Planetesimal Formation,” *ApJ*, vol. 969, no. 2, 130, p. 130, Jul. 2024.  DOI: 10.3847/1538-4357/ad47a2.
- 7 J. **Lim** and J. Cho, “Decay of Turbulence in Fluids with Polytrropic Equations of State,” *Journal of Korean Astronomical Society*, vol. 53, pp. 49–57, Apr. 2020.  DOI: 10.5303/JKAS.2020.53.2.49.
- 8 J. **Lim**, J. Cho, and H. Yoon, “Generation of Solenoidal Modes and Magnetic Fields in Turbulence Driven by Compressive Driving,” *ApJ*, vol. 893, no. 1, 75, p. 75, Apr. 2020.  DOI: 10.3847/1538-4357/ab8066.

## Posters

- 1 J. Lim and J. B. Simon, *Conditions for strong clumping by the streaming instability*, The Solar System in Context, University of Arizona, Tucson, AZ, Sep. 2025.
- 2 J. Lim and J. B. Simon, *Conditions for strong clumping by the streaming instability*, Gordon Research Conference, Mount Holyoke College, MA, Jun. 2025.
- 3 J. Lim and J. B. Simon, *Streaming instability and turbulence: Conditions for planetesimal formation*, Born in Fire, Santiago, Chile, Sep. 2024.
- 4 J. Lim and J. B. Simon, *Streaming instability and turbulence: Conditions for planetesimal formation*, New Heights in Planet Formation, Garching, Germany, Jul. 2024.
- 5 J. Lim and J. B. Simon, *Planetesimal formation by streaming instability: Effect of turbulence*, ERES VIII Symposium, Yale University, New Haven, CT, Jun. 2023.

## Presentations

- 1 J. Lim and J. B. Simon, *Simulating the streaming instability and planetesimal formation*, University of Nevada, Las Vegas, NV, Sep. 2025.
- 2 J. Lim and J. B. Simon, *Streaming instability and turbulence: Conditions for planetesimal formation*, TCAN Disk Hydro & Planet Formation, Tucson, AZ, Nov. 2023.
- 3 J. Lim and J. Cho, *Generation of magnetic fields in turbulence driven by compressive driving*, Magnetic Fields in the Universe 7, Quy Nhon, Vietnam, Feb. 2020.
- 4 J. Lim and J. Cho, *Generation of magnetic fields in turbulence driven by compressive driving*, Korean Astronomical Society Meeting, Seoul, South Korea, Oct. 2019.

## Professional and Committee Services

- 2025  External Reviewer for NASA’s Solar System Workings Program
- July 2018  LOC Member, 8th East-Asia School and Workshop on Laboratory Space and Astrophysical Plasmas

## Student Mentoring

- Fall 2022 - Spring 2024  Olivia Brouillette, former undergrad at ISU; now in graduate program at New Mexico State University

## Teaching Experience

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### Iowa State University, IA

Spring 2024        PHYS 132: General Physics II; Astro 150: Star, Galaxies, and Cosmology  
Fall 2023        PHYS 115: Physics for the Life Sciences  
Fall 2020        PHYS 111 Lab: General Physics

### Chungnam National University, Daejeon, South Korea

Spring 2019        300-level Galactic Astronomy

## References

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### Prof. Jacob B. Simon

Associate Professor  
Iowa State University  
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### Prof. Wladimir Lyra

Associate Professor  
New Mexico State University  
wlyra@nmsu.edu

### Prof. Steve Kawaler

Distinguished Professor  
Iowa State University  
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